

ASSIGNMENTS

Class-7

The Four Fundamental Subspaces

1. Describe (geometrically) the column space and row space of the following matrices:

i) $\begin{bmatrix} 1 & 2 \end{bmatrix}$

ii) $\begin{bmatrix} 1 & 2 & 0 \\ 1 & 2 & 0 \end{bmatrix}$

iii) $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$

iv) $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

v) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

v) $\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$

2. Find the column space and row space of $\begin{bmatrix} 2 & 3 & 1 \\ 1 & 2 & 1 \\ 3 & 5 & 2 \end{bmatrix}$

3. If A is any 5×5 invertible matrix, then what is the column space?
4. If A is any matrix of order 7×7 and the dimension of the $C(A^T)$ is 4, then what is the dimension of the $N(A)$?